SimWorkShare v0.4 - implementation-ready specification

Сравнение traditional_company, profit_sharing, employee_ownership_partial и worker_cooperative

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Версия документа: 0.4
Статус: implementation-ready specification
Назначение: экономическая имитационная модель для сравнения организационных систем компании.
Принцип нейтральности: модель не предполагает заранее превосходство кооператива, profit sharing или традиционной компании.

0. Анализ v0.3 и причины переработки

v0.3 была сфокусирована на сравнении fixed salary и profit sharing. Ее сильные элементы сохраняются: direct `behavior_case`, отделение P&L от cash-flow, AR/tax/bonus queues, restricted cash at accrual, common random numbers, paired deltas и sensitivity tests. Однако v0.3 не отвечает новой исследовательской цели, потому что не моделирует владение, участие работников в управлении, рост производственных возможностей, динамический headcount, ограничения внешнего капитала, концентрацию риска сотрудников и governance costs.

Проблема v0.3	Почему мешает v0.4	Решение v0.4
Объект сравнения - compensation scenarios	Новая цель сравнивает организационные системы	Ввести organizational_scenarios
employees_count постоянен	Нужны найм, увольнения, текучесть и итоговый размер	Dynamic workforce
Нет производственной мощности	Нельзя измерить долгосрочное развитие	productive_capacity_revenue_monthly и reinvestment lag
Profit sharing смешивает распределение и мотивацию	Ownership, profit distribution и governance должны быть отдельными механизмами	Независимые поля сценария
Нет governance costs/delay/quality	Участие в управлении может помогать или вредить	GovernanceModel
Главная метрика - owner cash	Новая цель исключает интересы основателя как критерий	Company/workforce/development/resilience metrics
Нет employee risk concentration	Доход и накопления могут зависеть от одной компании	employee_risk_concentration_index
Нет ограничений external capital	Они могут менять рост и устойчивость	external_capital_access_multiplier

1. Цель и границы модели

Цель SimWorkShare v0.4 - проверить гипотезу: при каких условиях коллективное владение компанией, участие сотрудников в управлении и распределение результатов повышают производительность, устойчивость и темпы развития компании по сравнению с традиционной организацией.

Модель рассчитывает численность сотрудников, найм, увольнения, производительность, выручку, расходы, прибыль, cash-flow, распределение прибыли, реинвестирование, денежный резерв, growth capacity, motivation assumptions, ownership retention effects, governance costs, decision quality/delay, free rider, fairness, employee risk concentration, capital constraints и реакцию на market shocks.

Модель не является юридической моделью кооператива, налоговой консультацией, бухгалтерским балансом полного стандарта, оценкой стоимости доли основателя или доказательством причинности.

2. Исследовательские вопросы и гипотезы

ID	Вопрос или гипотеза	Проверка
RQ1	При каком productivity uplift worker cooperative не уступает traditional company?	Break-even по sustainable_development_value_proxy и risk constraints
RQ2	Когда profit sharing дает эффект без ownership/governance?	Сравнить profit_sharing с traditional_company при behavior cases
RQ3	Компенсирует ли снижение текучести governance costs?	Paired deltas turnover, hiring costs, cash, capacity
RQ4	Когда fairness/free rider ухудшает outcome?	negative_fairness_free_rider sensitivity
RQ5	Как governance влияет на shock response и investment efficiency?	Decision quality/delay formulas

ID	Вопрос или гипотеза	Проверка
RQ6	Как risk concentration влияет на retention?	risk_concentration_turnover_sensitivity_annual_pp
RQ7	Какие системы устойчивее при шоках?	Shock survival rate and post-shock recovery
H1	Ownership может повысить retention	Только если задан отрицательный turnover delta
H2	Ownership может повысить productivity	Только если задан positive sensitivity
H3	Governance может повысить decision quality	Через explicit quality parameters
H4	Governance может замедлять решения	Через decision_delay_months
H5	Free rider может снизить productivity	Через free_rider_penalty
H6	Reinvestment повышает capacity	Через investment cash outflow and lag

3. Термины и определения

Термин	Определение
organizational_scenario	Полный набор параметров устройства компании: владение, управление, распределение результата, реинвестирование, доступ к капиталу
traditional_company	Фиксированная зарплата, centralized management, сотрудники не участвуют в собственности и profit distribution
profit_sharing	Фиксированная зарплата плюс cash distribution из результата; ownership и governance остаются традиционными
employee_ownership_partial	Промежуточная система частичного ownership и ограниченного governance voice
worker_cooperative	Коллективное владение сотрудниками, participatory governance, распределение части результата и обязательное reinvestment/reserve policy
productive_capacity_revenue_monthly	Максимальная месячная выручка, которую компания может произвести при достаточном спросе и труде
labor_revenue_capacity	Потенциальная выручка, ограниченная численностью и productivity
reinvestment	Денежный outflow на развитие, создающий capacity после лага
member_capital_account	Внутренний счет сотрудника, связанный с компанией и создающий employee risk concentration
common random numbers	Одинаковая внешняя траектория для сравниваемых сценариев в одном Monte Carlo run

4. Сценарии организационного устройства

Сценарий	Зарплата	Владение	Распределение результата	Управление	Реинвестирование	Доступ к капиталу
traditional_company	fixed	0	0	centralized	retained earnings policy	full
profit_sharing	fixed	0	cash distribution	centralized	company policy	full
employee_ownership_partial	fixed	partial	cash + optional member capital	partial voice	explicit	moderately constrained
worker_cooperative	fixed or stable base wage	collective	cash + member capital	participatory	explicit and protected	constrained

Механизмы не смешиваются. Формально `employee_ownership_fraction`, `employee_cash_distribution_rate`, `member_capital_allocation_rate`, `governance_participation_intensity` и `reinvestment_rate` являются независимыми параметрами.

5. Полный перечень входных параметров

Все `rate` задаются десятичной дробью: `0.10 = 10 percent`. Параметры с суффиксом `_pp` - absolute percentage points годовой ставки. Денежные величины выражаются в `simulation.currency`.

5.1. simulation

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
schema_version	string	version	0.4	0.4	Версия схемы конфигурации.
mode	enum	mode	deterministic	monte_carlo	monte_carlo
months	integer	months	1 to 240	240	Горизонт моделирования.
horizons_months	array[int]	months	1 to months	[60,120,240]	Горизонты отчета 5/10/20 лет.
runs	integer	runs	1 to 1000000	1000	Количество Monte Carlo прогонов.
random_seed	integer	seed	0 to 2^63-1	42	Воспроизводимость.
common_random_numbers	bool	bool	true	false	True
currency	string	currency	text	RUB	Валюта денежных параметров.
epsilon	number	currency/share	0 to 1e-3	1e-9	Числовая погрешность.
headcount_mode	enum	mode	fractional	integer_expected	integer_random
stop_after_bankruptcy	bool	bool	true	false	True

5.2. company_economics

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
initial_headcount	number/integer	people	>0	50	Начальная численность.
base_salary_per_employee_monthly	number/integer	currency/month/person	>=0	100000	Фиксированная зарплата.
salary_payroll_tax_rate	number/integer	share	0 to 1	0.0	Взносы/налоги работодателя на зарплату.
standard_hours_per_employee_month	number/integer	hours/person/month	1 to 300	160	Норма часов.
base_revenue_per_effective_employee_monthly	number/integer	currency/month/effective person	>=0	230000	Базовая выработка full-productivity employee.
initial_market_demand_monthly	number/integer	currency/month	>=0	11500000	Начальный внешний спрос.
initial_productive_capacity_revenue_monthly	number/integer	currency/month	>=0	13800000	Начальная производственная мощность.
fixed_costs_monthly	number/integer	currency/month	>=0	2000000	Постоянные расходы кроме payroll.
variable_cost_rate	number/integer	share of revenue	0 to 1	0.25	Переменные расходы.
profit_tax_rate	number/integer	share	0 to 1	0.2	Упрощенный налог на прибыль.
profit_tax_payment_lag_months	number/integer	months	0 to 24	1	Лаг оплаты налога.
cost_inflation_monthly	number/integer	monthly rate	-0.05 to 0.10	0.005	Инфляция зарплат и fixed costs.
capacity_depreciation_rate_monthly	number/integer	share/month	0 to 0.10	0.002	Устаревание производственных возможностей.
capacity_revenue_created_per_currency_invested	number/integer	currency monthly capacity / currency invested	>=0	0.08	Эффективность инвестиций в capacity.
investment_activation_lag_months	number/integer	months	0 to 60	3	Лаг ввода capacity.
required_cash_reserve_months	number/integer	months	0 to 24	2.0	Резерв обязательных платежей.
starting_cash	number/integer	currency	>=0	15000000	Денежный запас на старт.
opening_accounts_receivable	number/integer	currency	>=0	1725000	Дебиторка на старт.

5.3. market

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
market_process	number/enum/array	process	deterministic	bounded_lognormal	bounded_lognormal
market_growth_monthly	number/enum/array	monthly rate	-0.10 to 0.10	0.003	Средний рост спроса.
market_volatility_monthly	number/enum/array	std dev	0 to 1	0.08	Волатильность спроса.
market_factor_min	number/enum/array	multiplier	0 to 10	0.5	Нижняя граница market factor.
market_factor_max	number/enum/array	multiplier	0 to 10	1.8	Верхняя граница market factor.
seasonality_multipliers	number/enum/array	multiplier	12 values, 0 to 10	[1 x 12]	Сезонность спроса.
revenue_collection_rate_current_month	number/enum/array	share	0 to 1	0.85	Доля выручки, собранная cash в месяце.
accounts_receivable_lag_months	number/enum/array	months	0 to 24	1	Лаг сбора AR.
bad_debt_rate	number/enum/array	share	0 to 1	0.0	Безнадежная дебиторка.
shock_probability_monthly	number/enum/array	probability	0 to 1	0.03	Вероятность рыночного шока.
shock_revenue_multiplier	number/enum/array	multiplier	0 to 1	0.8	Базовый множитель выручки при шоке.
shock_cost_mean	number/enum/array	currency	>=0	0	Средний cost шока.
shock_cost_std	number/enum/array	currency	>=0	0	Стандартное отклонение shock cost.
cash_collection_stress_multiplier	number/enum/array	multiplier	0 to 1	0.9	Снижение collection rate при шоке.
labor_market_factor	number/enum/array	multiplier	0 to 5	1.0	Множитель текучести.
credit_market_factor	number/enum/array	multiplier	0 to 5	1.0	Доступность внешнего капитала.

5.4. workforce

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
base_turnover_rate_annual	number/enum/array	annual share	0 to 1	0.2	Базовая годовая текучесть.
min_turnover_rate_annual	number/enum/array	annual share	0 to 1	0.03	Нижняя граница.
max_turnover_rate_annual	number/enum/array	annual share	0 to 1	0.6	Верхняя граница.

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
turnover_random_mode	number/enum/array	mode	deterministic	binomial	deterministic
high_performer_share	number/enum/array	share	0 to 1	0.2	Доля сильных сотрудников для risk indicator.
recruiting_cost_per_hire	number/enum/array	currency/hire	>=0	50000	Стоимость найма.
onboarding_cost_per_hire	number/enum/array	currency/hire	>=0	50000	Адаптация/обучение.
manager_time_cost_per_hire	number/enum/array	currency/hire	>=0	25000	Время менеджмента.
exit_admin_cost_per_leaver	number/enum/array	currency/leaver	>=0	10000	Админ. стоимость ухода.
lost_productivity_cost_per_leaver	number/enum/array	currency/leaver	>=0	0	Явная стоимость потерь; default 0 чтобы не double count.
severance_cost_per_layoff	number/enum/array	currency/layoff	>=0	100000	Стоимость сокращения.
ramp_duration_months	number/enum/array	months	0 to 24	3	Период выхода на продуктивность.
ramp_productivity_multipliers	number/enum/array	multiplier	0 to 1	[0.5,0.75,0.9]	Продуктивность новых сотрудников.
max_hires_per_month_rate	number/enum/array	share of headcount	0 to 1	0.1	Ограничение найма.
max_layoffs_per_month_rate	number/enum/array	share of headcount	0 to 1	0.1	Ограничение сокращений.
layoff_trigger_cash_ratio	number/enum/array	ratio	0 to 10	0.5	Порог cash stress.
leaver_paid_fraction_of_month	number/enum/array	share	0 to 1	0.5	Доля месяца, оплаченная уходящим.
new_hire_paid_fraction_of_month	number/enum/array	share	0 to 1	0.5	Доля месяца, оплаченная новым.
min_productivity_uplift	number/enum/array	uplift	-1 to 1	-0.15	Нижняя граница uplift.
max_productivity_uplift	number/enum/array	uplift	-1 to 1	0.2	Верхняя граница uplift.
turnover_productivity_penalty_per_annual_turnover	number/enum/array	uplift loss	0 to 10	0.1	Потеря productivity от excess turnover.
target_staffing_buffer	number/enum/array	multiplier	0 to 3	1.05	Запас headcount под спрос.
max_cash_share_for_hiring	number/enum/array	share	0 to 1	0.25	Доля excess cash на hiring.

5.5. employee_risk

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
employee_external_savings_proxy_per_employee	number	currency/person	>=0	600000	Proxy внешних накоплений сотрудника.
employment_dependence_index	number	index	0 to 1	1.0	Зависимость занятости от компании.
risk_weight_variable_income	number	weight	0 to 1	0.4	Вес переменного дохода.
risk_weight_member_capital	number	weight	0 to 1	0.4	Вес member capital.
risk_weight_employment_dependence	number	weight	0 to 1	0.2	Вес зависимости занятости.

5.6. financing

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
base_credit_line	number/enum	currency	>=0	0	Доступная кредитная линия.
debt_interest_rate_annual	number/enum	annual rate	0 to 1	0.18	Стоимость долга.
scheduled_principal_payment_monthly	number/enum	currency/month	>=0	0	Плановое погашение principal.
external_growth_capital_limit_monthly	number/enum	currency/month	>=0	0	Максимальный внешний капитал на развитие.
external_capital_type	number/enum	type	debt	non_dilutive_grant	debt
distribution_payroll_tax_rate	number/enum	share	0 to 1	0.0	Налоги/взносы на employee distribution.
distribution_tax_deductible_share	number/enum	share	0 to 1	1.0	Доля distribution, уменьшающая taxable profit.
employee_distribution_payout_lag_months	number/enum	months	0 to 24	1	Лаг выплаты employee distribution.
member_capital_redemption_lag_months	number/enum	months	0 to 120	24	Лаг выплаты member capital при уходе.
member_capital_redemption_fraction	number/enum	share	0 to 1	1.0	Доля member capital к redemption.

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
_on_exit					
reserve_release_rate_on_stress	number/enum	share/month	0 to 1	0.5	Release restricted reserve under stress.

5.7. organizational_scenarios

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
name	number/string/enum/array/null	id	unique snake_case	required	Имя сценария.
system_type	number/string/enum/array/null	type	traditional_company	profit_sharing	employee_ownership_partial
employee_ownership_fraction	number/string/enum/array/null	share	0 to 1	scenario-specific	Экономическое участие сотрудников.
employee_cash_distribution_rate	number/string/enum/array/null	share	0 to 1	scenario-specific	Cash distribution сотрудникам.
member_capital_allocation_rate	number/string/enum/array/null	share	0 to 1	scenario-specific	Начисление на внутренние счета.
reinvestment_rate	number/string/enum/array/null	share	0 to 1	scenario-specific	Доля результата на развитие.
organizational_reserve_rate	number/string/enum/array/null	share	0 to 1	scenario-specific	Резерв устойчивости.
external_distribution_rate	number/string/enum/array/null	share	0 to 1	0	Cash outflow вне компании; не target metric.
result_hurdle_monthly	number/string/enum/array/null	currency/month	>=0	0	Результат, не подлежащий распределению.
allocation_priority	number/string/enum/array/null	order	valid allocation ids	required	Очередь распределения результата.
distribution_rule	number/string/enum/array/null	rule	none	equal_per_capita	contribution_weighted
contribution_measurement_quality	number/string/enum/array/null	index	0 to 1	scenario-specific	Качество измерения индивидуального вклада.
peer_monitoring_effectiveness	number/string/enum/array/null	index	0 to 1	scenario-specific	Снижение free-rider через peer monitoring.
transparency_index	number/string/enum/array/null	index	0 to 1	scenario-specific	Прозрачность формулы и финансов.
employment_stabilization_preference	number/string/enum/array/null	index	0 to 1	scenario-specific	Склонность избегать layoffs.
external_capital_access_multiplier	number/string/enum/array/null	multiplier	0 to 5	scenario-specific	Ограничение/преимущество внешнего капитала.
profit_distribution_period_months	number/string/enum/array/null	months	1 to 12	1	Период начисления distribution.
max_distribution_per_employee_period	number/string/enum/array/null	currency/person/period	>=0	null	None
behavior_case_refs	number/string/enum/array/null	ids	non-empty	required	Какие behavior cases прогонять.

5.8. governance

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
governance_participation_intensity	number	varies	see spec	0.8	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
base_governance_hours_per_employee_month	number	varies	see spec	4.0	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
fixed_governance_hours_monthly	number	varies	see spec	40.0	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
governance_cash_cost_fixed_monthly	number	varies	see spec	100000	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
governance_cash_cost_per_employee_monthly	number	varies	see spec	2000	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
decision_complexity_index	number	varies	see spec	1.2	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
base_decision_delay_months	number	varies	see spec	0.25	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
delay_per_participation_months	number	varies	see spec	0.5	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
local_autonomy_index	number	varies	see spec	0.6	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
decentralization_speed_gain_months	number	varies	see spec	0.2	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
governance_capability_index	number	varies	see spec	1.0	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
information_sharing_quality	number	varies	see spec	0.8	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
trust_index	number	varies	see spec	0.7	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
quality_gain_from_participation	number	varies	see spec	0.03	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
coordination_loss_from_participation	number	varies	see spec	0.02	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
conflict_loss_sensitivity	number	varies	see spec	0.02	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
decision_delay_quality_loss	number	varies	see spec	0.01	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
decision_quality_min	number	varies	see spec	0.7	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
decision_quality_max	number	varies	see spec	1.2	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
shock_mitigation_sensitivity	number	varies	see spec	0.15	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.
shock_delay_amplification	number	varies	see spec	0.03	Параметр модели управления;

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
					влияет на hours, cost, delay, quality, shock response или investment efficiency.
investment_efficiency_sensitivity	number	varies	see spec	0.15	Параметр модели управления; влияет на hours, cost, delay, quality, shock response или investment efficiency.

5.9. behavior_cases

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
base_productivity_uplift_direct	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
ownership_productivity_sensitivity	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
profit_distribution_productivity_sensitivity	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
governance_voice_productivity_sensitivity	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
base_turnover_delta_annual_pp	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
ownership_retention_delta_annual_pp_per_full_ownership	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
profit_distribution_retention_delta_annual_pp_per_10pp	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
governance_retention_delta_annual_pp_per_full_participation	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
fairness_base	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
transparency_to_fairness	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
equal_distribution_fairness_effect	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
contribution_based_distribution_fairness_effect	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
pay_dispersion_fairness_penalty	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
unpaid_governance_burden_penalty	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
zero_distribution_fairness_penalty	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
fairness_productivity_sensitivity	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
fairness_turnover_sensitivity_annual_	number	varies	see spec	0.0	Сценарное допущение поведения;

Имя	Тип	Ед.	Диапазон	Default	Экономический смысл
pp					в no_effect равно нейтральному значению.
free_rider_base_penalty	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
free_rider_size_exponent	number	varies	see spec	0.5	Сценарное допущение поведения; в no_effect равно нейтральному значению.
free_rider_reference_headcount	number	varies	see spec	50	Сценарное допущение поведения; в no_effect равно нейтральному значению.
free_rider_max_size_multiplier	number	varies	see spec	3.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
risk_concentration_turnover_sensitivity_annual_pp	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
income_volatility_turnover_sensitivity_annual_pp	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.
high_performer_attrition_delta_pp	number	varies	see spec	0.0	Сценарное допущение поведения; в no_effect равно нейтральному значению.

6. Состояние модели на начало месяца

Группа	State variables на начало месяца t
EnvironmentState	month, market_trend_t, market_factor_t, seasonality_multiplier_t, shock_happened_t, shock_cost_t, collection_rate_multiplier_t, labor_market_factor_t, credit_market_factor_t
WorkforceState	headcount_begin_t, ramp_cohorts_begin_t[k], full_productivity_headcount_begin_t, income_history_12m, zero_distribution_streak, last_turnover_rate_annual
GovernanceState	governance_capability_index_t, trust_index_t, information_sharing_quality_t, decision_delay_months_t, decision_quality_multiplier_t
ProductionState	productive_capacity_revenue_monthly_begin_t, capacity_addition_queue, capacity_depreciation_rate_monthly
FinancialState	cash_total_begin_t, restricted_distribution_cash_begin_t, restricted_reserve_cash_begin_t, debt_balance_begin_t, ar_queue, tax_payable_queue, employee_distribution_payable_queue, member_capital_redemption_queue
MemberCapitalState	member_capital_accounts_total_begin_t, member_capital_per_employee_average_begin_t
RiskState	min_unrestricted_cash_to_date, bankruptcy_absorbed_flag, months_since_first_shock, unpaid_mandatory_obligations_to_date

Definitions:

```
restricted_cash_total_t = restricted_distribution_cash_t + restricted_reserve_cash_t
unrestricted_cash_t = cash_total_t - restricted_cash_total_t
```

7. Точный порядок расчетов внутри месяца

1. Если `bankruptcy_absorbed_flag=true` и `stop_after_bankruptcy=true`, записать inactive month.
2. Получить `EnvironmentMonth[t]` из common environment path.
3. Ввести due capacity additions.
4. Рассчитать governance hours, costs, decision delay и decision quality.
5. Рассчитать fairness, free rider, employee risk concentration и behavioral effects.
6. Рассчитать voluntary turnover.
7. Рассчитать layoffs на основе beginning cash stress и employment stabilization policy.
8. Рассчитать desired headcount, hiring need, hiring affordability и hires.
9. Обновить ramp cohorts для production текущего месяца.

10. Рассчитать effective employees и productivity multiplier.
11. Рассчитать market demand, labor revenue capacity, productive capacity limit и revenue.
12. Рассчитать accrual operating costs.
13. Начислить AR и собрать cash из текущей выручки и прошлой AR.
14. Рассчитать due obligations: taxes, employee distributions, member redemptions, debt service.
15. Оплатить обязательные cash obligations.
16. При необходимости высвободить restricted reserve по stress policy.
17. При необходимости привлечь credit line для mandatory cash gap.
18. Рассчитать operating profit before allocation.
19. Рассчитать cash-safe base for allocations.
20. По `allocation_priority` распределить positive result.
21. Рассчитать taxable profit, profit tax accrual и поставить tax в queue.
22. Поставить employee distribution в payable queue и зарезервировать cash.
23. Провести cash reinvestment и optional external growth capital.
24. Поставить capacity additions в queue с лагом.
25. Рассчитать closing cash, restricted cash, debt, member capital, headcount.
26. Рассчитать risk flags.
27. Сохранить `MonthlyResult`.

8. Формулы для каждого этапа

8.1. Environment path

```
market_trend_t = market_trend_{t-1} * (1 + market_growth_monthly)
raw_market_factor_t = exp((-0.5 * sigma_m^2) + sigma_m * z_market_t)
market_factor_t = clamp(raw_market_factor_t, market_factor_min, market_factor_max)
shock_happened_t ~ Bernoulli(shock_probability_monthly)
collection_rate_multiplier_t = cash_collection_stress_multiplier if shock_happened_t else 1
```

In deterministic mode: `market_factor_t=1`, no random shocks except explicit schedule, `runs=1`.

8.2. Governance cost and decision quality

```
governance_hours_t = headcount_begin_t * governance_participation_intensity * base_governance_hours_per_employee_month + fixed_governance_hours_monthly
governance_admin_equivalent_employees_t = governance_hours_t / standard_hours_per_employee_month
governance_cash_cost_t = governance_cash_cost_fixed_monthly + headcount_begin_t * governance_cash_cost_per_employee_monthly

decision_delay_months_t = max(0,
  base_decision_delay_months
  + decision_complexity_index * governance_participation_intensity * delay_per_participation_months / governance_capability_index
  - local_autonomy_index * decentralization_speed_gain_months
)

decision_quality_raw_t = 1
  + quality_gain_from_participation * governance_participation_intensity * information_sharing_quality
  - coordination_loss_from_participation * governance_participation_intensity * decision_complexity_index
  - conflict_loss_sensitivity * (1 - trust_index)
  - decision_delay_quality_loss * decision_delay_months_t

decision_quality_multiplier_t = clamp(decision_quality_raw_t, decision_quality_min, decision_quality_max)
```

8.3. Fairness

```
distribution_rule_effect_t =
  0 for none
```

```
  equal_distribution_fairness_effect for equal_per_capita
  contribution_based_distribution_fairness_effect * contribution_measurement_quality for contribution_weighted
  0.5 * equal_distribution_fairness_effect + 0.5 * contribution_based_distribution_fairness_effect * contribution_measurement_quality for hybrid

governance_burden_share_t = governance_hours_t / max(epsilon, headcount_begin_t * standard_hours_per_employee_month)

fairness_index_t = clamp(
  fairness_base + transparency_to_fairness * transparency_index + distribution_rule_effect_t
  - pay_dispersion_fairness_penalty * pay_dispersion_index_t
  - unpaid_governance_burden_penalty * governance_burden_share_t
  - zero_distribution_fairness_penalty * zero_distribution_streak_t,
  -1, 1
)
```

Default `pay_dispersion_index_t=0` unless explicitly configured.

8.4. Free rider

```
collective_distribution_exposure_t = employee_cash_distribution_rate + member_capital_allocation_rate
equal_distribution_factor_t = 1 for equal_per_capita, 0.5 for hybrid, 0 otherwise
size_factor_t = min(free_rider_max_size_multiplier, (headcount_begin_t / free_rider_reference_headcount) ^ free_rider_size_exponent)
monitoring_reduction_t = peer_monitoring_effectiveness * contribution_measurement_quality
free_rider_penalty_t = free_rider_base_penalty * collective_distribution_exposure_t * equal_distribution_factor_t * size_factor_t * (1 - monitoring_reduction_t)
```

8.5. Employee risk concentration

```
variable_income_share_12m_t = employee_cash_distribution_paid_12m_t / max(epsilon, fixed_salary_paid_12m_t + employee_cash_distribution_paid_12m_t)
member_capital_per_employee_t = member_capital_accounts_total_begin_t / max(epsilon, headcount_begin_t)
capital_concentration_t = member_capital_per_employee_t / max(epsilon, member_capital_per_employee_t + employee_external_savings_proxy_per_employee)
employee_risk_concentration_index_t = clamp(
  risk_weight_variable_income * variable_income_share_12m_t
  + risk_weight_member_capital * capital_concentration_t
  + risk_weight_employment_dependence * employment_dependence_index,
  0, 1
)
```

8.6. Productivity uplift

```
ownership_salience_t = employee_ownership_fraction * transparency_index
profit_distribution_salience_t = employee_cash_distribution_rate / 0.10
governance_salience_t = governance_participation_intensity

motivation_uplift_raw_t = base_productivity_uplift_direct
  + ownership_productivity_sensitivity * ownership_salience_t
  + profit_distribution_productivity_sensitivity * profit_distribution_salience_t
  + governance_voice_productivity_sensitivity * governance_salience_t

fairness_productivity_effect_t = fairness_productivity_sensitivity * fairness_index_t
turnover_excess_t = max(0, turnover_rate_annual_t - base_turnover_rate_annual)
turnover_productivity_loss_t = turnover_productivity_penalty_per_annual_turnover * turnover_excess_t

productivity_uplift_t = clamp(
  motivation_uplift_raw_t + fairness_productivity_effect_t
  + governance_voice_productivity_sensitivity * max(0, decision_quality_multiplier_t - 1)
  - free_rider_penalty_t - turnover_productivity_loss_t,
  min_productivity_uplift, max_productivity_uplift
)
productivity_multiplier_t = 1 + productivity_uplift_t
```

8.7. Turnover

```
fairness_turnover_delta_t = - fairness_turnover_sensitivity_annual_pp * fairness_index_t
risk_turnover_delta_t = risk_concentration_turnover_sensitivity_annual_pp * employee_risk_concentration_index_t
income_volatility_turnover_delta_t = income_volatility_turnover_sensitivity_annual_pp * income_volatility_index_12m_t
ownership_turnover_delta_t = ownership_retention_delta_annual_pp_per_full_ownership * ownership_salience_t
distribution_turnover_delta_t = profit_distribution_retention_delta_annual_pp_per_10pp * profit_distribution_salience_t
governance_turnover_delta_t = governance_retention_delta_annual_pp_per_full_participation * governance_salience_t
```

```
turnover_rate_annual_t = clamp(
  base_turnover_rate_annual * labor_market_factor_t
  + base_turnover_delta_annual_pp
  + ownership_turnover_delta_t + distribution_turnover_delta_t + governance_turnover_delta_t
  + fairness_turnover_delta_t + risk_turnover_delta_t + income_volatility_turnover_delta_t,
  min_turnover_rate_annual, max_turnover_rate_annual
)
turnover_rate_monthly_t = 1 - (1 - turnover_rate_annual_t)^(1/12)
voluntary_leavers_t = headcount_begin_t * turnover_rate_monthly_t  # deterministic mode
```

8.8. Layoffs and hiring

```
required_cash_reserve_begin_t = required_cash_reserve_months * (salary_costs_reference_monthly_t + fixed_costs_reference_monthly_t + scheduled_debt_service_reference_monthly_t)
begin_cash_stress_ratio_t = unrestricted_cash_begin_t / max(epsilon, required_cash_reserve_begin_t)
cash_stress_layoff_pressure_t = max(0, layoff_trigger_cash_ratio - begin_cash_stress_ratio_t) / max(epsilon, layoff_trigger_cash_ratio)
distress_layoff_rate_t = max_layoffs_per_month_rate * cash_stress_layoff_pressure_t * (1 - employment_stabilization_preference)
distress_layoffs_t = min(headcount_begin_t - voluntary_leavers_t, (headcount_begin_t - voluntary_leavers_t) * distress_layoff_rate_t)

target_revenue_for_staffing_t = min(market_demand_forecast_t * target_staffing_buffer, productive_capacity_revenue_monthly_begin_t)
desired_headcount_t = target_revenue_for_staffing_t / (base_revenue_per_effective_employee_monthly * max(0.10, productivity_multiplier_t))
gross_hiring_need_t = max(0, desired_headcount_t - headcount_after_exits_t)
hire_capacity_t = max_hires_per_month_rate * max(1, headcount_begin_t)
hiring_cash_available_t = max(0, unrestricted_cash_begin_t - required_cash_reserve_begin_t) * max_cash_share_for_hiring
affordable_hires_t = hiring_cash_available_t / max(epsilon, recruiting_cost_per_hire + onboarding_cost_per_hire + manager_time_cost_per_hire)
hires_t = min(gross_hiring_need_t, hire_capacity_t, affordable_hires_t)
headcount_end_t = headcount_begin_t - voluntary_leavers_t - distress_layoffs_t + hires_t
```

8.9. Effective employees and revenue

```
effective_ramping_employees_t = sum_k ramp_cohort_after_exits_t[k] * ramp_productivity_multipliers[k]
effective_new_hires_t = hires_t * ramp_productivity_multipliers[0]
effective_employees_t = max(0, full_productivity_headcount_after_exits_t + effective_ramping_employees_t + effective_new_hires_t - governance_admin_equivalent_employees_t)

base_shock_loss_t = 1 - shock_revenue_multiplier
shock_loss_multiplier_t = clamp(1 - shock_mitigation_sensitivity * (decision_quality_multiplier_t - 1) + shock_delay_amplification * decision_delay_months_t, 0, 3)
effective_shock_revenue_multiplier_t = 1 - clamp(base_shock_loss_t * shock_loss_multiplier_t, 0, 1)

market_demand_t = initial_market_demand_monthly * market_trend_t * seasonality_multiplier_t * market_factor_t * effective_shock_revenue_multiplier_t
labor_revenue_capacity_t = effective_employees_t * base_revenue_per_effective_employee_monthly * productivity_multiplier_t
productive_capacity_limit_t = productive_capacity_revenue_monthly_begin_t
revenue_t = max(0, min(market_demand_t, labor_revenue_capacity_t, productive_capacity_limit_t))
```

8.10. Costs, cash collection and mandatory payments

```
paid_employees_t = headcount_begin_t - leaver_paid_fraction_of_month * voluntary_leavers_t - leaver_paid_fraction_of_month * distress_layoffs_t + new_hire_paid_fraction_of_month *
hires_t
salary_cost_t = paid_employees_t * base_salary_per_employee_monthly * cumulative_cost_inflation_t
salary_payroll_tax_t = salary_cost_t * salary_payroll_tax_rate
fixed_costs_t = fixed_costs_monthly * cumulative_cost_inflation_t
variable_costs_t = revenue_t * variable_cost_rate
hiring_cost_t = hires_t * (recruiting_cost_per_hire + onboarding_cost_per_hire + manager_time_cost_per_hire)
exit_cost_t = voluntary_leavers_t * (exit_admin_cost_per_leaver + lost_productivity_cost_per_leaver)
layoff_cost_t = distress_layoffs_t * severance_cost_per_layoff
operating_costs_before_allocation_t = salary_cost_t + salary_payroll_tax_t + fixed_costs_t + variable_costs_t + hiring_cost_t + exit_cost_t + layoff_cost_t + governance_cash_cost_t +
shock_cost_accrual_t

cash_collected_current_t = revenue_t * clamp(revenue_collection_rate_current_month * collection_rate_multiplier_t, 0, 1)
new_ar_t = revenue_t * (1 - effective_collection_rate_t) * (1 - bad_debt_rate)
cash_collected_from_revenue_t = cash_collected_current_t + ar_queue.amount_due(t)

mandatory_cash_payments_t = salary_cost_t + salary_payroll_tax_t + fixed_costs_t + variable_costs_t + turnover_and_workforce_cost_t + governance_cash_cost_t + shock_cost_accrual_t +
taxes_due_t + employee_distribution_due_t + member_capital_redemption_due_t + interest_expense_t + principal_payment_t
cash_after_mandatory_before_financing_t = cash_total_begin_t + cash_collected_from_revenue_t - mandatory_cash_payments_t
credit_draw_for_liquidity_t = min(max(0, -cash_after_mandatory_before_financing_t), credit_headroom_t)
cash_after_mandatory_t = cash_after_mandatory_before_financing_t + credit_draw_for_liquidity_t
```

8.11. Result allocation, tax, reinvestment and capacity growth

```
operating_profit_before_allocation_t = revenue_t - operating_costs_before_allocation_t
profit_before_tax_before_distribution_t = operating_profit_before_allocation_t - interest_expense_t
positive_result_base_t = max(0, profit_before_tax_before_distribution_t - result_hurdle_monthly)

cash_safe_allocation_budget_t = max(0, unrestricted_cash_before_allocations_t - required_cash_reserve_t - tax_reserve_estimate_t)

raw_employee_cash_distribution_gross_t = employee_cash_distribution_rate * positive_result_base_t
raw_member_capital_allocation_t = member_capital_allocation_rate * positive_result_base_t
raw_reinvestment_t = reinvestment_rate * positive_result_base_t
raw_organizational_reserve_t = organizational_reserve_rate * positive_result_base_t
raw_external_distribution_t = external_distribution_rate * positive_result_base_t
```

Validation: `employee_cash_distribution_rate + member_capital_allocation_rate + reinvestment_rate + organizational_reserve_rate + external_distribution_rate <= 1`.

For each item in `allocation_priority`:

```
actual_i_t = min(raw_i_t, remaining_allocation_budget_t / cash_multiplier_i)
remaining_allocation_budget_t -= actual_i_t * cash_multiplier_i
```

Cash multipliers: employee cash distribution uses `1 + distribution_payroll_tax_rate`; other allocation types use `1`.

```
deductible_employee_distribution_t = actual_employee_cash_distribution_gross_t * distribution_tax_deductible_share
taxable_profit_t = max(0, profit_before_tax_before_distribution_t - deductible_employee_distribution_t)
profit_tax_accrual_t = taxable_profit_t * profit_tax_rate

actual_reinvestment_total_t = cash_reinvestment_paid_t + external_growth_capital_draw_t
investment_efficiency_multiplier_t = clamp(1 + investment_efficiency_sensitivity * (decision_quality_multiplier_t - 1), 0, 3)
capacity_added_by_investment_t = actual_reinvestment_total_t * capacity_revenue_created_per_currency_invested * investment_efficiency_multiplier_t
effective_investment_lag_t = investment_activation_lag_months + round(decision_delay_months_t)
capacity_addition_queue.add(t + effective_investment_lag_t, capacity_added_by_investment_t)
```

8.12. Closing state and bankruptcy

```
restricted_distribution_cash_close_t = restricted_distribution_cash_begin_t - employee_distribution_due_t + restricted_distribution_cash_new_t
restricted_reserve_cash_close_t = restricted_reserve_cash_after_release_t + actual_organizational_reserve_t
cash_total_close_t = cash_total_after_growth_capital_t
unrestricted_cash_close_t = cash_total_close_t - restricted_distribution_cash_close_t - restricted_reserve_cash_close_t

reserve_breach_flag_t = unrestricted_cash_close_t < required_cash_reserve_t
cash_gap_flag_t = cash_total_close_t < 0
liquidity_deficit_flag_t = unpaid_mandatory_obligations_t > epsilon
bankruptcy_flag_t = liquidity_deficit_flag_t OR debt_balance_close_t > credit_line_limit_t + epsilon
```

9. Правила распределения прибыли и реинвестирования

Распределение выполняется только из `positive_result_base_t` и только в пределах `cash_safe_allocation_budget_t`. Положительная accounting profit не означает автоматическую выплату.

Сценарий	Default policy
traditional_company	employee_cash_distribution_rate=0, member_capital_allocation_rate=0, governance=0, reinvestment_rate=0.15, organizational_reserve_rate=0.05
profit_sharing	employee_cash_distribution_rate=0.10, ownership=0, governance=0, reinvestment_rate=0.10, reserve=0.05
employee_ownership_partial	ownership=0.30, cash distribution=0.10, member capital=0.05, governance=0.30, reinvestment=0.15, reserve=0.05
worker_cooperative	ownership=1.00, cash distribution=0.15, member capital=0.05, governance=0.80, reinvestment=0.25, reserve=0.10

10. Поведенческие механизмы

Поведение задается через explicit behavior cases. Модель не считает motivation effect установленным фактом.

Case	Смысл
no_effect	Ownership, profit sharing и governance не дают productivity/retention эффекта
retention_only	Влияет только на удержание
moderate_positive	Умеренный положительный uplift/retention
governance_costly	Participation создает costs/delay без productivity gain
negative_fairness_free_rider	Fairness/free rider ухудшают productivity и turnover
risk_concentration_negative	Variable income и member capital повышают turnover

Положительный механизм	Ограничивающий механизм
Ownership может повысить commitment	Concentrated risk может повысить turnover
Profit distribution может повысить effort	Variable income volatility может снизить attractiveness
Governance voice может повысить качество решений	Governance может замедлить решения и снизить productive hours
Equal distribution может повысить cohesion	Equal distribution может демотивировать high performers
Member capital может удерживать сотрудников	Member capital создает redemption liabilities и employee risk
Reinvestment повышает capacity	Reinvestment снижает short-term cash
Employment stabilization снижает layoffs	Может увеличить bankruptcy risk при downturn

11. Модель управления и стоимости принятия решений

Governance влияет на модель по четырем каналам: снижает effective employees через governance hours; увеличивает cash costs; замедляет решения через decision delay; меняет quality через decision quality multiplier. Для `traditional_company` положительное качество не предполагается автоматически. Для `worker_cooperative` высокая participation не создает uplift в `no_effect`.

12. Рыночная среда и случайные процессы

Все внешние процессы генерируются один раз на `run` и переиспользуются всеми сценариями при `common_random_numbers=true`.

Stream	Использование
market_stream	market factor
shock_stream	Bernoulli shocks
shock_cost_stream	shock cost
labor_stream	stochastic turnover, если включен
credit_stream	stochastic credit market factor, если включен

13. Денежные очереди и обязательства

Queue	Что хранит	Cash effect
ar_queue	дебиторка к сбору	cash inflow in due month
tax_payable_queue	начисленный profit tax	cash outflow in due month
employee_distribution_payable_queue	начисленные выплаты сотрудникам	cash restricted at accrual, cash outflow at payment
member_capital_redemption_queue	выплаты внутренних счетов при уходе	cash outflow in due month
capacity_addition_queue	будущий ввод мощности	capacity increase in due month

Priority cash payments: payroll, fixed costs, variable costs, workforce costs, governance cash costs, shock costs, taxes due, employee distributions due, member capital redemptions due, interest, principal, then new discretionary allocations.

14. Риски и условия банкротства

Flag	Формула
reserve_breach_flag	unrestricted_cash_close_t < required_cash_reserve_t
cash_gap_flag	cash_total_close_t < 0
liquidity_deficit_flag	unpaid_mandatory_obligations_t > epsilon
credit_limit_breach_flag	debt_balance_close_t > credit_line_limit_t + epsilon
bankruptcy_flag	liquidity_deficit_flag OR credit_limit_breach_flag
employee_distribution_cut_flag	raw distribution > actual distribution
reinvestment_underfunded_flag	raw reinvestment > actual reinvestment total

After bankruptcy and if `stop_after_bankruptcy=true`, the scenario becomes inactive, generates no revenue and performs no hiring.

15. Выходные показатели и формат отчета

15.1. Company metrics

Метрика	Формула
monthly_revenue	revenue_t
cumulative_revenue	sum_t revenue_t

Метрика	Формула
operating_profit_before_allocation	revenue_t - operating_costs_before_allocation_t
net_profit_after_tax_and_employee_distribution	see tax formula
productive_capacity_revenue_monthly	closing capacity
capacity_growth_rate	capacity_close / capacity_start - 1
final_headcount	headcount_end_T
revenue_cagr	(revenue_last_12m / revenue_first_12m)^(12/(T-12)) - 1
capacity_cagr	(capacity_T / capacity_0)^(12/T) - 1
cash_end_total	cash_total_close_T
cash_end_unrestricted	unrestricted_cash_close_T
min_unrestricted_cash	min_t unrestricted_cash_close_t
liquidity_deficit_probability	share of runs with any liquidity deficit
bankruptcy_probability	share of runs with any bankruptcy
shock_survival_rate	share of shock runs with no bankruptcy within 12 months after shock

15.2. Workforce and employee metrics

Метрика	Формула
productivity_per_employee	revenue_t / max(epsilon, paid_employees_t)
turnover_rate_annual_average	average monthly annualized turnover
voluntary_leavers_total	sum_t voluntary_leavers_t
layoffs_total	sum_t distress_layoffs_t
hires_total	sum_t hires_t
hiring_and_onboarding_costs_total	sum_t hiring_cost_t
average_employee_income_monthly	(salary_paid + distribution_paid) / average_headcount
employee_cash_distribution_total	sum_t employee_cash_distribution_paid_t
member_capital_accounts_total	closing member capital
employee_income_volatility	std dev monthly employee income
risk_adjusted_employee_income	average_income - volatility_penalty_lambda * income_volatility
employee_risk_concentration_index_average	average over months

15.3. Development metrics

```
reinvestment_total_cash = sum_t cash_reinvestment_paid_t
external_growth_capital_total = sum_t external_growth_capital_draw_t
actual_reinvestment_total = sum_t actual_reinvestment_total_t
reinvestment_underfunding_rate = sum(raw_reinvestment - actual_reinvestment_total) / sum(raw_reinvestment)
productive_capacity_added_total = sum_t capacity_added_by_investment_t
capacity_replacement_cost_proxy = productive_capacity_revenue_monthly / max(epsilon, capacity_revenue_created_per_currency_invested)
sustainable_development_value_proxy = cash_end_unrestricted + capacity_replacement_cost_proxy - debt_balance - unpaid_obligations - member_capital_redemption_due
```

`Sustainable_development_value_proxy` не является стоимостью доли основателя.

15.4. Summary output fields

```
scenario, system_type, behavior_case, market_case, horizon_months,
median_cumulative_revenue, p10_cumulative_revenue,
median_operating_profit, median_net_profit,
median_productivity_per_employee, turnover_rate_annual_average,
hiring_and_onboarding_costs_total, average_employee_income_monthly,
risk_adjusted_employee_income, employee_cash_distribution_total,
member_capital_accounts_total, reinvestment_total_cash,
productive_capacity_growth_rate, cash_end_total_median,
cash_end_unrestricted_p10, min_unrestricted_cash_p10,
liquidity_deficit_probability, bankruptcy_probability,
shock_survival_rate, final_headcount_median,
revenue_cagr_median, capacity_cagr_median,
employee_risk_concentration_index_average, classification, assumption_flags
```

15.5. Monthly CSV fields

```
run, month, scenario, system_type, behavior_case, market_case, active_company_flag,
market_factor, shock_happened, effective_collection_rate, headcount_begin,
voluntary_leavers, layoffs, hires, headcount_end, effective_employees,
productivity_uplift, productivity_multiplier, governance_hours,
governance_admin_equivalent_employees, governance_cash_cost, decision_delay_months,
decision_quality_multiplier, fairness_index, free_rider_penalty,
employee_risk_concentration_index, market_demand, labor_revenue_capacity,
productive_capacity_revenue_monthly, revenue, salary_cost, fixed_costs,
variable_costs, turnover_and_workforce_cost, shock_cost,
operating_profit_before_allocation, interest_expense,
profit_before_tax_before_distribution, positive_result_base,
cash_collected_current, cash_collected_from_ar, mandatory_cash_payments,
cash_after_mandatory, credit_draw_for_liquidity,
employee_cash_distribution_accrued, employee_cash_distribution_paid,
member_capital_allocation, member_capital_redemption_due,
reinvestment_cash_paid, external_growth_capital_draw,
organizational_reserve_allocation, profit_tax_accrual, taxes_paid,
cash_total_close, restricted_distribution_cash_close,
restricted_reserve_cash_close, unrestricted_cash_close, debt_balance_close,
required_cash_reserve, reserve_breach_flag, cash_gap_flag,
liquidity_deficit_flag, bankruptcy_flag
```

16. Парное сравнение сценариев

For each run `r`, scenario `s`, reference `b`:

```
paired_delta_X[r,s,b] = X[r,s] - X[r,b]
```

Default reference is `traditional_company`. Additional reference `profit_sharing` isolates incremental ownership/governance effects. Report median, P10, P90, probability positive and probability negative. A scenario cannot be classified as development-dominant if bankruptcy or liquidity deficit probability is worse than reference by more than tolerance.

17. Monte Carlo и sensitivity analysis

Minimum Monte Carlo: `runs>=1000`, `common_random_numbers=true`, horizons include 60 and 120 months. Publication-quality stress report should use `runs>=10000`.

Mandatory sensitivity grid includes productivity uplift, ownership productivity sensitivity, retention effect, governance intensity/hours, decision quality, free rider, fairness, risk concentration, external capital access, reinvestment rate, cash reserve, shock probability, shock severity and collection rate.

Break-even motivation effect:

```
BE_productivity_uplift_s = inf { u |
  median(paired_delta_sustainable_development_value_proxy(u)) >= 0
  AND bankruptcy_probability_s(u) <= bankruptcy_probability_b + tolerance
  AND liquidity_deficit_probability_s(u) <= liquidity_deficit_probability_b + tolerance
}
```

If no value exists inside `break_even_uplift_range`, return null and flag `no_break_even_in_tested_range`.

18. Проверяемые инварианты

Инвариант	Формула
Headcount identity	headcount_end = headcount_begin - voluntary_leavers - layoffs + hires
Non-negative headcount	headcount_end >= -epsilon
Revenue bounds	revenue <= market_demand, revenue <= labor_revenue_capacity, revenue <= productive_capacity_limit
Cash identity	cash_close = cash_begin + collections + credit_draws + external_capital_inflows - mandatory_payments - reinvestment_cash - external_distribution - external_capital_spent
Restricted cash identity	restricted_cash = restricted_distribution_cash + restricted_reserve_cash
Unrestricted cash identity	unrestricted_cash = cash_total - restricted_cash
Debt identity	debt_close = debt_begin - principal_paid + credit_draw_for_liquidity + external_growth_debt_draw
Capacity identity	capacity_close = capacity_begin * (1 - depreciation) + capacity_additions_due
Member capital identity	member_capital_close = member_capital_begin + allocation - redemption_accrual
Allocation sum	sum(policy rates) <= 1
Cash-safe allocation	sum(actual_allocations * cash_multiplier) <= cash_safe_allocation_budget + epsilon

Инвариант	Формула
No automatic motivation	In no_effect, structural ownership/profit/governance cannot create productivity/retention effect
Bankruptcy absorbing	If bankruptcy at t, active flag false for all future months when enabled

19. Валидация конфигурации

Strict validation must reject unknown fields, duplicate JSON fields, NaN/Infinity, missing required fields, invalid enums, negative money where disallowed, rates outside [0,1], horizons greater than months, duplicate scenario names, unknown behavior/scenario refs, invalid allocation priority items and allocation rate sums above 1.

Normalization rules: annual rates are not overwritten; monthly equivalents are derived. Null caps mean no cap. Empty `behavior_case_refs` is an error. In deterministic mode `runs` must be `1`. `ramp_productivity_multipliers.length` must equal `ramp_duration_months` unless duration is 0.

Warnings: possible turnover double count, inconsistent scenario label, high redemption liquidity risk, external distribution not being a success metric, simplified tax model.

20. Набор обязательных тестов и эталонных примеров

The canonical test manifest is provided in `required_tests_v0_4.json`. Required tests include baseline sanity, headcount identity, no-effect control, profit sharing without ownership, governance cost, cash-safe distribution, distribution reserve, member capital redemption queue, reinvestment capacity lag, external capital constraint, common random numbers, shock response quality, bankruptcy absorbing behavior, strict unknown-field rejection, duplicate JSON key rejection and v0.3 compatibility tests.

21. Ограничения модели и недопустимые интерпретации

The model must not be interpreted as proof that one organizational system is universally better. Behavioral effects are assumptions. The workforce is aggregated. Tax logic is simplified. The model is not a full balance sheet. Member capital is simplified. Governance is stylized. Market paths are stress tools. External capital is simplified. Sustainable development value proxy is not equity value.

Недопустимые выводы: `worker_cooperative` всегда лучше; `traditional_company` всегда эффективнее; motivation effect доказан моделью; модель оценила стоимость доли основателя; результат можно применять без sensitivity analysis.

22. План миграции с v0.3

Сохраняются: monthly discrete simulation, deterministic/Monte Carlo, random seed, common random numbers, paired deltas, P&L/cash separation, AR queue, tax payable queue, employee distribution payable queue, restricted cash at accrual, cash-safe distribution base, behavior cases, no-effect and negative controls, sensitivity outputs, monthly CSV, validation and sanity tests.

Deprecated as primary metrics: `owner_distributable_cash`, `owner_dividend_policy`, `fixed_raise_same_expected_cost` as main comparator, constant `employees_count`, `demand_cap_multiplier` as sole capacity constraint.

New modules: `WorkforceDynamics`, `ProductionCapacity`, `GovernanceModel`, `OwnershipAndMemberCapital`, `RiskConcentration`, `ExternalCapitalAccess`, `ResultAllocationPolicy`, `ShockResponse`, `DevelopmentMetrics`, `CompatibilityV03`.

`profit_share_equal_10` maps to `profit_sharing` with ownership=0, governance=0, cash distribution=0.10, member capital=0, distribution rule `equal_per_capita`, period 1 month, payout lag 1 month.

Compatibility mode must reproduce `fixed_only`, profit share 5/10/15, above-hurdle variants, monthly/quarterly/annual periods, v0.3 cash-safe accrual, restricted cash at accrual, AR/tax/bonus queues and v0.3 risk flags.

See `migration_map_v0_3_to_v0_4.csv` for field-level mapping.

23. Критерии готовности реализации

Implementation is ready when: strict validation passes; all organizational scenarios are implemented; ownership, distribution and governance are independent; no-effect does not create uplift; headcount and cash identities hold; P&L and cash are separated; distributions are cash-safe and restricted at accrual; reinvestment is cash outflow that creates capacity after lag; governance costs/delay/quality work by formula; free rider/fairness/risk concentration are explicit; external capital constraints affect liquidity and growth; shocks use common random numbers; paired deltas are calculated per run; Monte Carlo is reproducible by seed; deterministic mode passes sanity tests; sensitivity and break-even analyses work; v0.3 compatibility tests pass; reports exclude founder-share value as a target metric; limitations and assumption flags are printed.

Appendix A. Architecture

Модуль	Ответственность
SimulationEngine	runs, seed, common random numbers, horizons, scenario matrix
ScenarioExpander	matrix organizational_scenario x behavior_case x market_case
EnvironmentPath	рынок, инфляция, шоки, трудовой рынок, кредитный рынок

Модуль	Ответственность
WorkforceDynamics	headcount, hires, leavers, layoffs, ramp-up, текучесть
BehaviorMechanisms	motivation assumptions, fairness, free rider, risk concentration, retention
GovernanceModel	governance hours, cost, delay, decision quality
ProductionModel	labor capacity, productive capacity, demand, revenue
CompanyEconomics	revenue, costs, operating profit, net profit
ResultAllocationPolicy	distribution, reinvestment, reserves, member capital
CashFlowModel	cash, restricted cash, AR, taxes, distributions, debt
FinancingModel	credit line, liquidity draw, external growth capital
Metrics	monthly, cumulative, paired deltas, percentiles, risk metrics
SensitivityRunner	grids, tornado, break-even motivation effect
Validation	config validation, invariants, unit and sanity tests
CompatibilityV03	v0.3 reproduction mode

Appendix B. Full JSON configuration example


```json
{
  "schema_version": "0.4",
  "config_validation": {
    "reject_unknown_fields": true,
    "reject_duplicate_fields": true,
    "allow_name_normalization": false,
    "reject_nan_and_infinity": true
  },
  "simulation": {
    "mode": "monte_carlo",
    "months": 240,
    "horizons_months": [
      60,
      120,
      240
    ],
    "runs": 1000,
    "random_seed": 42,
    "common_random_numbers": true,
    "currency": "RUB",
    "epsilon": 1e-09,
    "headcount_mode": "fractional",
    "stop_after_bankruptcy": true
  },
  "units": {
    "money": "RUB nominal unless simulation.currency is changed",
    "rates": "decimal share, 0.10 means 10 percent",
    "percentage_points": "absolute annual rate change, -0.03 means -3 pp",
    "time_step": "one month",
    "company_economics.initial_headcount": "people",
    "company_economics.base_salary_per_employee_monthly": "RUB per person per month",
    "company_economics.base_revenue_per_effective_employee_monthly": "RUB revenue per effective employee per month",
    "company_economics.initial_market_demand_monthly": "RUB revenue demand per month",
    "company_economics.initial_productive_capacity_revenue_monthly": "RUB revenue capacity per month",
    "company_economics.fixed_costs_monthly": "RUB per month",
    "company_economics.capacity_revenue_created_per_currency_invested": "RUB monthly revenue capacity per RUB invested",
    "market.market_growth_monthly": "monthly rate",
    "market.market_volatility_monthly": "monthly standard deviation",
    "workforce.base_turnover_rate_annual": "annual share",
    "workforce.ramp_productivity_multipliers": "productivity multiplier by tenure month",
    "employee_risk.employee_external_savings_proxy_per_employee": "RUB per employee",
    "financing.base_credit_line": "RUB",
    "organizational_scenarios.employee_cash_distribution_rate": "share of positive result base",
    "organizational_scenarios.member_capital_allocation_rate": "share of positive result base",
    "organizational_scenarios.reinvestment_rate": "share of positive result base",
    "organizational_scenarios.governance.governance_participation_intensity": "0 to 1 index"
  },
```

```
"company_economics": {
  "initial_headcount": 50,
  "base_salary_per_employee_monthly": 100000,
  "salary_payroll_tax_rate": 0.0,
  "standard_hours_per_employee_month": 160,
  "base_revenue_per_effective_employee_monthly": 230000,
  "initial_market_demand_monthly": 11500000,
  "initial_productive_capacity_revenue_monthly": 13800000,
  "fixed_costs_monthly": 2000000,
  "variable_cost_rate": 0.25,
  "profit_tax_rate": 0.2,
  "profit_tax_payment_lag_months": 1,
  "cost_inflation_monthly": 0.005,
  "capacity_depreciation_rate_monthly": 0.002,
  "capacity_revenue_created_per_currency_invested": 0.08,
  "investment_activation_lag_months": 3,
  "required_cash_reserve_months": 2.0,
  "starting_cash": 15000000,
  "opening_accounts_receivable": 1725000
},
"market": {
  "market_process": "bounded_lognormal",
  "market_growth_monthly": 0.003,
  "market_volatility_monthly": 0.08,
  "market_factor_min": 0.5,
  "market_factor_max": 1.8,
  "seasonality_multipliers": [
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0,
    1.0
  ],
  "revenue_collection_rate_current_month": 0.85,
  "accounts_receivable_lag_months": 1,
  "bad_debt_rate": 0.0,
  "shock_probability_monthly": 0.03,
  "shock_revenue_multiplier": 0.8,
  "shock_cost_mean": 0,
  "shock_cost_std": 0,
  "cash_collection_stress_multiplier": 0.9,
  "labor_market_factor": 1.0,
  "credit_market_factor": 1.0
},
"workforce": {
  "base_turnover_rate_annual": 0.2,
  "min_turnover_rate_annual": 0.03,
  "max_turnover_rate_annual": 0.6,
  "turnover_random_mode": "deterministic",
  "high_performer_share": 0.2,
  "recruiting_cost_per_hire": 50000,
  "onboarding_cost_per_hire": 50000,
  "manager_time_cost_per_hire": 25000,
  "exit_admin_cost_per_leaver": 10000,
  "lost_productivity_cost_per_leaver": 0,
  "severance_cost_per_layoff": 100000,
  "ramp_duration_months": 3,
  "ramp_productivity_multipliers": [
    0.5,
```

```
    0.75,
    0.9
  ],
  "max_hires_per_month_rate": 0.1,
  "max_layoffs_per_month_rate": 0.1,
  "layoff_trigger_cash_ratio": 0.5,
  "leaver_paid_fraction_of_month": 0.5,
  "new_hire_paid_fraction_of_month": 0.5,
  "min_productivity_uplift": -0.15,
  "max_productivity_uplift": 0.2,
  "turnover_productivity_penalty_per_annual_turnover": 0.1,
  "target_staffing_buffer": 1.05,
  "max_cash_share_for_hiring": 0.25
},
"employee_risk": {
  "employee_external_savings_proxy_per_employee": 600000,
  "employment_dependence_index": 1.0,
  "risk_weight_variable_income": 0.4,
  "risk_weight_member_capital": 0.4,
  "risk_weight_employment_dependence": 0.2
},
"financing": {
  "base_credit_line": 0,
  "debt_interest_rate_annual": 0.18,
  "scheduled_principal_payment_monthly": 0,
  "external_growth_capital_limit_monthly": 0,
  "external_capital_type": "debt",
  "distribution_payroll_tax_rate": 0.0,
  "distribution_tax_deductible_share": 1.0,
  "employee_distribution_payout_lag_months": 1,
  "member_capital_redemption_lag_months": 24,
  "member_capital_redemption_fraction_on_exit": 1.0,
  "reserve_release_rate_on_stress": 0.5
},
"behavior_cases": {
  "no_effect": {
    "base_productivity_uplift_direct": 0.0,
    "ownership_productivity_sensitivity": 0.0,
    "profit_distribution_productivity_sensitivity": 0.0,
    "governance_voice_productivity_sensitivity": 0.0,
    "base_turnover_delta_annual_pp": 0.0,
    "ownership_retention_delta_annual_pp_per_full_ownership": 0.0,
    "profit_distribution_retention_delta_annual_pp_per_10pp": 0.0,
    "governance_retention_delta_annual_pp_per_full_participation": 0.0,
    "fairness_base": 0.0,
    "transparency_to_fairness": 0.0,
    "equal_distribution_fairness_effect": 0.0,
    "contribution_based_distribution_fairness_effect": 0.0,
    "pay_dispersion_fairness_penalty": 0.0,
    "unpaid_governance_burden_penalty": 0.0,
    "zero_distribution_fairness_penalty": 0.0,
    "fairness_productivity_sensitivity": 0.0,
    "fairness_turnover_sensitivity_annual_pp": 0.0,
    "free_rider_base_penalty": 0.0,
    "free_rider_size_exponent": 0.5,
    "free_rider_reference_headcount": 50,
    "free_rider_max_size_multiplier": 3.0,
    "risk_concentration_turnover_sensitivity_annual_pp": 0.0,
    "income_volatility_turnover_sensitivity_annual_pp": 0.0,
    "high_performer_attrition_delta_pp": 0.0
  },
  "retention_only": {
    "base_productivity_uplift_direct": 0.0,
    "ownership_productivity_sensitivity": 0.0,
    "profit_distribution_productivity_sensitivity": 0.0,
    "governance_voice_productivity_sensitivity": 0.0,
```

```
    "base_turnover_delta_annual_pp": 0.0,
    "ownership_retention_delta_annual_pp_per_full_ownership": -0.04,
    "profit_distribution_retention_delta_annual_pp_per_10pp": -0.01,
    "governance_retention_delta_annual_pp_per_full_participation": -0.01,
    "fairness_base": 0.0,
    "transparency_to_fairness": 0.1,
    "equal_distribution_fairness_effect": 0.0,
    "contribution_based_distribution_fairness_effect": 0.0,
    "pay_dispersion_fairness_penalty": 0.0,
    "unpaid_governance_burden_penalty": 0.0,
    "zero_distribution_fairness_penalty": 0.0,
    "fairness_productivity_sensitivity": 0.0,
    "fairness_turnover_sensitivity_annual_pp": 0.02,
    "free_rider_base_penalty": 0.0,
    "free_rider_size_exponent": 0.5,
    "free_rider_reference_headcount": 50,
    "free_rider_max_size_multiplier": 3.0,
    "risk_concentration_turnover_sensitivity_annual_pp": 0.0,
    "income_volatility_turnover_sensitivity_annual_pp": 0.0,
    "high_performer_attrition_delta_pp": 0.0
  },
  "moderate_positive": {
    "base_productivity_uplift_direct": 0.0,
    "ownership_productivity_sensitivity": 0.03,
    "profit_distribution_productivity_sensitivity": 0.005,
    "governance_voice_productivity_sensitivity": 0.01,
    "base_turnover_delta_annual_pp": 0.0,
    "ownership_retention_delta_annual_pp_per_full_ownership": -0.04,
    "profit_distribution_retention_delta_annual_pp_per_10pp": -0.01,
    "governance_retention_delta_annual_pp_per_full_participation": -0.01,
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    "equal_distribution_fairness_effect": 0.02,
    "contribution_based_distribution_fairness_effect": 0.02,
    "pay_dispersion_fairness_penalty": 0.0,
    "unpaid_governance_burden_penalty": 0.05,
    "zero_distribution_fairness_penalty": 0.02,
    "fairness_productivity_sensitivity": 0.02,
    "fairness_turnover_sensitivity_annual_pp": 0.02,
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    "free_rider_size_exponent": 0.5,
    "free_rider_reference_headcount": 50,
    "free_rider_max_size_multiplier": 3.0,
    "risk_concentration_turnover_sensitivity_annual_pp": 0.01,
    "income_volatility_turnover_sensitivity_annual_pp": 0.01,
    "high_performer_attrition_delta_pp": 0.0
  },
  "governance_costly": {
    "base_productivity_uplift_direct": 0.0,
    "ownership_productivity_sensitivity": 0.0,
    "profit_distribution_productivity_sensitivity": 0.0,
    "governance_voice_productivity_sensitivity": 0.0,
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}
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Appendix C. Self-check

Проверка	Статус
Нет ли противоречий между разделами?	Явных противоречий нет
Определены ли все используемые переменные?	Да: через parameters, state или formulas

Проверка	Статус
Однозначен ли порядок расчета месяца?	Да: раздел 7
Можно ли реализовать модель без дополнительных экономических решений?	Да: formulas, defaults, ranges and priorities are specified
Нейтральна ли модель?	Да: no-effect prevents automatic motivation effect
Можно ли проверить формулы тестами?	Да: invariants and required test manifest are specified